

UNIVERSITEIT VAN PRETORIA / UNIVERSITY OF PRETORIA

DEPARTEMENT WISKUNDE EN TOEGEPASTE WISKUNDE
DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

CALCULUS WTW 158
EXAMINATION / EKSAMEN
Interne Eksaminator/ Internal Examiner:
Dr AF Harding

(Section A)
Quiz not
included

Eksterne Eksaminator/ External Examiner:
Prof JC Engelbrecht

November 2002
Tyd/Time: 30 min
Punte/Marks: 20

VAN/SURNAME: _____

VOORNAME/FIRST NAMES: _____

STUDENTENOMMER/STUDENT NUMBER: _____

HANDTEKENING/SIGNATURE: _____

**PUNTE
MARKS**

LEES DIE VOLGENDE INSTRUKSIES

1. Die vraestel bestaan uit bladsye 1 tot 3. Kontroleer of jou vraestel volledig is.
2. As jy meer as die beskikbare ruimte vir 'n antwoord nodig het, gebruik dan die laaste bladsy en trek 'n raam daar om.

READ THE FOLLOWING INSTRUCTIONS

1. The paper consists of pages 1 to 3. Check whether your paper is complete.
2. If you need more than the available space for an answer, use the last page and please indicate it clearly by *framing* it.

Question 1 / Vraag 1 [5]

Consider the function / Beskou die funksie

$$f(x) = x + \cos 2x$$

on the interval $[0, 1]$ / op die interval $[0, 1]$.

1.1 Find the interval(s) on which f increases and the interval(s) on which f decreases /
Bepaal die interval(le) waarop f styg en die interval(le) waarop f daal.

1.2 Determine the turning point(s) of the function on the interval and classify it as a local maximum or a local minimum. /

Bepaal die draaipunt(e) van die funksie op die interval en klassifiseer dit as 'n lokale maksimum of 'n lokale minimum.

1.3 Use the Closed Interval method to determine the absolute maximum and absolute minimum of the function on the interval. / Gebruik die Geslote Interval metode om die absolute maksimum en absolute minimum van die funksie op die interval te bepaal.

Question 2 / Vraag 2 [5]

2.1 For what value of x is the tangent to the curve $3x = 2 - e^{2x}$ parallel to the line $2x - y = 4$?

Vir watter waarde van x is die raaklyn aan die kromme $3x = 2 - e^{2x}$ ewewydig aan die lyn $2x - y = 4$?

2.2 Determine $\frac{dy}{dt}$ if / Bepaal $\frac{dy}{dt}$ as

$$y = \frac{T}{\rho g} \cosh\left(\frac{\rho g a}{T}\right), \quad \rho, g, a \text{ constants / konstantes}$$

Question 3 / Vraag 3 [4]

Determine the following integrals: / Bepaal die volgende integrale:

3.1 $\int \sin 3x \sqrt{1 + \cos 3x} dx$

3.2 $\int \sin^3 2\theta d\theta$

Question 4 / Vraag 4 [3]

Determine the angle between the two intersecting lines $\langle x, y, z \rangle = \langle 4, 1, 8 \rangle + t \langle 1, -2, 3 \rangle$ and $\langle x, y, z \rangle = \langle 4, 1, 8 \rangle + t \langle 2, 1, 2 \rangle$.

Bepaal die hoek tussen die snyende lyne $\langle x, y, z \rangle = \langle 4, 1, 8 \rangle + t \langle 1, -2, 3 \rangle$ en $\langle x, y, z \rangle = \langle 4, 1, 8 \rangle + t \langle 2, 1, 2 \rangle$

Question 5 / Vraag 5 [3]

Find the equation of the plane through the point $(1, 0, 1)$ and parallel to the plane $2x + 4y - z = 7$.

Gee die vergelyking van die vlak deur die punt $(1, 0, 1)$ en ewewydig aan die vlak $2x + 4y - z = 7$.